

OpenClaw AI Weather Station Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the AI weather station project itself.

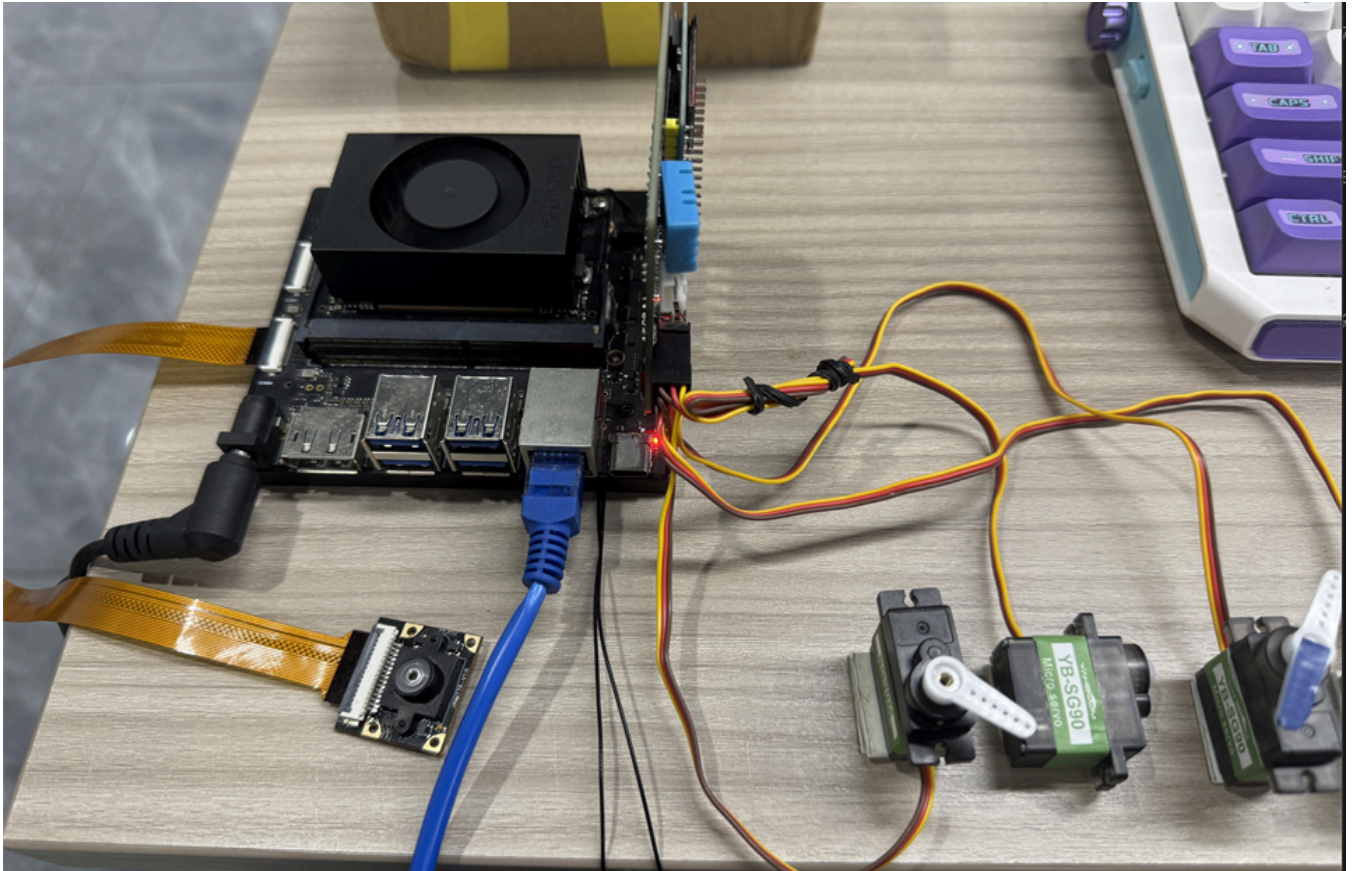
1. Tutorial Objectives

This article demonstrates OpenClaw's capabilities in desktop weather station scenarios through temperature/humidity collection, information display, and AI explanation. After completing this tutorial, readers can:

- Understand how to combine DHT, OLED, and RGB to create a local AI weather station
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Complete four project scenarios: environment collection, AI broadcast, weather card, status alert

2. Hardware Preparation

- OpenClaw mainboard
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB light bead module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Temperature and humidity reading command: `/home/jetson/hardware_ha1_demo/dht`
- OLED control command: `/home/jetson/hardware_ha1_demo/oled`
- RGB control command: `/home/jetson/hardware_ha1_demo/rgb`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

The current set of tutorials focuses on "local environmental weather station", which is based on DHT indoor temperature and humidity data for display and explanation;

4. Test Project Basic Capabilities

Before starting this project, please complete one `openclaw_tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter AI weather station specific testing.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then test the following types of commands:

- Read current temperature and humidity
- Display weather card on OLED
- If environment is normal, light green
- Give me a brief environment broadcast

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote environment viewing, classroom demonstrations	Suitable for remote inquiry "how is the environment now"	Demonstrate "remote weather station assistant"
TUI	Local debugging, threshold verification	Easiest to troubleshoot sensor and display logic	First run through collection chain
Voice code	Voice broadcast, desktop assistant	Closer to daily usage experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
[播报] 开始播报 tui 回复
```

5.1 whatsapp Solution

For whatsapp connection and robot creation steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. After creation, you can remotely view environmental status in whatsapp, for example:

- How is the indoor environment now? If it's too hot, remind me

20:34



< 2



+86 153 3885 7526 (你)

给自己发消息

...training for me...

20:33 ✓✓

How is the indoor environment now? If it is too hot, remind me.

20:34 ✓✓

Current indoor environment readings:



Temperature: 29.7°C (85.5°F)



Humidity: 54%

Assessment: The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

Recommendation:

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓



5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Get Beijing's current weather and display it on OLED

```
File Edit Tabs Help
Schedule,, but no pet food was found. Want to try again with dog food in the
frame? [0 1 4]
[9 8 4]

Get Beijing's current weather and display it on OLED

Beijing weather displayed on OLED! [0 1 4]
[3 2 4]

Current Weather in Beijing:
- Condition: Sunny *
- Temperature: 29°C (85°F)
- Humidity: 58%
- Wind: 18 km/h from South
- Feels like: 27°C

OLED Display shows:
- Line 1: Beijing
- Line 2: Sunny
- Line 3: Temp 29C Hum 58%

The weather card is now showing on your OLED screen! ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
68k / 1.0m (7%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- AI weather station

```
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写
```

```
[你] Ai weather station.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200
```

```
[OpenClaw] ## 🌤️AI Weather Station Complete!
```

```
**Local Sensor Readings:**
```

```
-
-           **Indoor Temperature:** 30°C (86°F)
- **Indoor Humidity:** 51%
-           **Outdoor Weather (Shenzhen):**
- **Condition:** Partly cloudy 🌤️
-
-           **Outdoor Temperature:** 28°C (82°F)
- **Feels Like:** 35°C (95°F)
-
-           **Outdoor Humidity:** 84%
- **Wind:** 9 km/h from South
- **Rain
-           Chance:** 72% today
```

```
**3-Day Forecast:**
```

```
- **Today (May 14):** High
-           28°C, Low 23°C - Patchy rain nearby
- **Tomorrow (May 15):** High
-           26°C, Low 23°C - Moderate rain expected
```

6. Project Implementation

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Scenario One: Environment Collection

Experimental Objective

Read the local environment's temperature and humidity, completing the most basic data sampling for the weather station.

Effect Description

OpenClaw will call DHT to obtain current environmental data, and return results to the user, forming a real-time "weather reading" of the indoor environment.

Example Instructions

- Read current environmental temperature and humidity
- What is the current indoor temperature and humidity
- Help me check the current weather data

6.2 Scenario Two: AI Broadcast

Experimental Objective

Let OpenClaw not only report numbers, but also summarize environmental status like a weather broadcaster.

Effect Description

OpenClaw can summarize "26 degrees, 57% humidity" into more natural words, such as "environment is relatively comfortable", "somewhat stuffy", "air is slightly dry", etc.

Example Instructions

- Give me a brief environment broadcast
- Is the current environment comfortable
- Make a weather summary based on current readings

6.3 Scenario Three: OLED Weather Card

Experimental Objective

Display the local temperature and humidity results as a more "weather card" style page.

Effect Description

Besides pure text results, OLED can also arrange location, status, temperature, and humidity into a concise card, making the entire project more like a displayable desktop device.

Example Instructions

- Display an indoor weather card on OLED
- Get current weather for XX location, and display on OLED

6.4 Scenario Four: RGB Status Alert

Experimental Objective

Use RGB colors to quickly express current environmental status.

Effect Description

For example, when environment is comfortable, light green; when too hot, light yellow; when abnormal, light red. This way, even without looking at text, you can quickly know the current environmental status.

Example Instructions

- If environment is normal, light green
- If temperature is high, light yellow
- If abnormal, light red blinking